



Social Dimensions in Urban Waste Management: Preliminary Study for the Design of a Waste Information System in Gorontalo City

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Abstract: Urban waste management requires a data-driven approach to understand community characteristics as a basis for designing public service information systems. This study aims to analyze the social dimensions of household waste management in Gorontalo City as a basis for the pre-design stage of the waste management information system. The research design used a cross-sectional survey of 400 households selected through stratified random sampling in nine subdistricts. Data were collected through a structured questionnaire covering sociodemographic characteristics, types of waste produced, and waste disposal behavior. The analysis was conducted descriptively and inferentially using the Chi-square and Cramer's V tests. The results show that household waste is dominated by organic waste (73%) and plastic (24%). The most common disposal behavior is through government transportation services (46.5%) and public trash bins (31.8%), while burning waste (15.0%) and disposal into rivers/open spaces (6.3%) are still found. Although there were minor variations in the contingency table between socio-demographic groups, the Chi-square test results showed that gender, age, and education were not significantly related to waste type and disposal behavior ($p > 0.05$). This indicates that waste management behavior is relatively homogeneous across all social groups. These findings reinforce the need for a universal service approach to waste management and provide an empirical basis for the development of the Gorontalo City SIMS, which focuses on improving service access, reporting disposal points, public education, and the integration of waste banks and city waste facilities.

Keywords: Waste management; Community behavior; Chi-square; Crosstab; Waste information system; Gorontalo

1 Introduction

Waste management is a crucial challenge for cities in Indonesia, including Gorontalo City. Population increase, urbanization, and lifestyle changes have caused the volume of waste to increase significantly every year. Data from Waste Management of the Indonesian Ministry of Environment noted that in 2024, Gorontalo City's waste generation reached 142 tons per day, in line with this, the Gorontalo City Environment Office in January 2025 claimed a significant increase in the amount of waste production reaching 180 tons per day [1–3].

The increase in waste volume certainly causes public green areas in the city of Gorontalo to experience a worrying year-on-year reduction, in line with population growth [4, 5]. The expansion of residential areas and economic activities has reduced the extent of natural vegetation, thereby decreasing environmental carrying capacity [6, 7]. One of the most striking consequences is the volume of waste that is not optimally managed. The piles of garbage seen in various corners of the city not only mar the city's appearance but also the government's efforts to make Gorontalo a smart city. This also causes air pollution due to the unpleasant smell from decaying organic waste. Domestic waste that is mixed without sorting worsens air quality and the surrounding environment [8, 9]. If not addressed promptly with an integrated management strategy, this issue can threaten public health and reduce the quality of life in the future.

Although various efforts have been made, waste management in Indonesia still faces numerous obstacles. Data shows that household waste is still managed in an environmentally unfriendly manner, such as being burned, disposed of in drainage systems, or thrown away carelessly. This condition reflects the low awareness and behavior of the community in sustainable waste management [10]. In the city of Gorontalo, as one of the developing cities in Indonesia, waste management has also become an urgent issue [11]. The absence of specific data regarding community behavior in waste disposal and the types of waste generated daily makes it difficult to formulate effective and targeted policies.

Most of the previous studies focused more on the technical aspects of waste management, such as processing technology and infrastructure, as well as the economic aspects, such as management costs and efficiency. However, social aspects, particularly people's waste disposal behavior and the type of waste generated based on socio-demographic characteristics such as gender, age, and education level, have received less attention. In fact, understanding this social dimension is very important for designing more effective and sustainable interventions and policies. Without understanding people's behavior and characteristics, waste management efforts risk not achieving optimal results.

In addition, efforts to improve waste management at the city level are currently also directed at utilizing digital technology through the development of waste service information systems. Various cities in Indonesia have begun to implement waste reporting systems, disposal point mapping, transportation schedule integration, and application-based Waste Bank services. However, the effectiveness of these information systems is highly dependent on understanding the behavior, habits, and preferences of the community as the main users. Therefore, a social analysis of waste types and community behavior in disposing of waste is a crucial first step before designing the Gorontalo City waste management information system. This approach ensures that the system design is in line with local needs, characteristics, and culture, making it easier for residents to accept and adopt.

Thus, this study aims not only to map the social dimensions of household waste management in Gorontalo City, but also to provide an empirical basis for the development of an effective waste information system based on the local context. A comprehensive social analysis is expected to provide input for the design of public service features, digital education, and mapping of waste facilities in this region.

Based on this problem, the research aims to analyze the behavior of Gorontalo City residents in disposing of daily waste. Then identify the types of waste produced by the community every day and examine the relationship between waste disposal behavior and the type of waste produced with the socio-demographic characteristics of the community, namely gender, age, and education level. The results of this study will serve as a reference in designing the core features of the waste management application, including the integration of transportation services, reporting of disposal points, access to the Waste Bank, community education, and mapping of municipal waste facilities. With an evidence-based approach, the resulting system is expected to support more effective and participatory waste management in Gorontalo City.

2 Literature Review

Waste management in Indonesia's urban areas faces complex challenges that are not only technical, but also social. Population increase, urbanization, and lifestyle changes have caused the volume of waste to increase significantly every year. However, without a deep understanding of people's behavior, technical efforts in waste management are often ineffective. According to previous research, household socio-economic characteristics, such as education level and number of family members, have a significant effect on waste management behavior. In addition, environmental risk awareness, which is influenced by knowledge and religious values, also plays an important role in determining people's behavior towards waste management [12, 13].

Various studies have shown that sociodemographic factors such as age, gender, and education level influence community waste-disposal behavior. The study by Han et al. [14] shows that age, gender, and individual perceptions are significantly related to waste separation behavior. For example, individuals with a positive perception of waste management tend to be more active in separating household waste. Additionally, Eshete et al. [15] and Wang et al. [16] stated that knowledge, attitudes, and the availability of waste disposal facilities are closely related to waste disposal behavior. Respondents with positive knowledge and attitudes, as well as access to adequate waste disposal facilities, tend to exhibit better waste disposal behavior.

People's knowledge and attitude towards the environment play a crucial role in waste management. Research by Sharma et al. [17] showed that there is a significant relationship between knowledge and attitude with waste disposal behavior. People with good knowledge about the negative impact of waste on the environment tend to have more responsible attitudes and behaviors in disposing of waste. Similarly, a study by Poyai et al. [18] identified that knowledge, attitude, action, facility support, and janitor support significantly influenced waste management behavior. Action is the dominant factor with a POR value of 8.826, indicating that real action in waste management is strongly influenced by individual knowledge and attitude.

3 Methods

This study uses a quantitative approach with a survey method to obtain an overview of the behavior of Gorontalo City residents in waste management, in terms of social dimensions such as gender, age, and education level. Data was collected through distributing structured questionnaires to 400 respondents spread across 9 sub-districts and covering 50 villages in Gorontalo City. The 400 respondents were obtained from a sampling process using the Slovin formula from a population of 159,339 in Gorontalo City with a margin of error of 5%. The selection of respondents was done proportionally to ensure a balanced representation of all administrative areas of the city, taking into account the population density in each kelurahan. The sampling technique used was stratified random sampling, to ensure representation of the main demographic groups that are the focus of the research.

The questionnaire instrument was designed to assess the daily waste management conditions as well as the habits or patterns of waste disposal by the community. The collected data was then analyzed using descriptive statistical techniques, crosstab analysis, and chi-square to examine the relationship between sociodemographic variables and waste management behavior characteristics. In addition, the validity and reliability of the instrument were tested in a pilot test conducted outside the main sample to ensure that the instrument is able to measure the intended variables consistently and accurately. The data collection process was conducted during March to April 2025, with the involvement of field enumerators who had been given prior training to be able to explain the questions in the questionnaire appropriately to respondents. All data obtained was processed using statistical software (SPSS) to ensure the accuracy of the results and ease of interpretation.

The researchers also added that, given the cross-sectional nature of the data and the fairly detailed categories, some tables had cells with expected counts <5 . Therefore, the inferential results were interpreted without claiming causality, and when not significant, they were combined with trend pattern readings to conclude the homogeneity of behavior relevant to the pre-design of a universal service-based waste management information system.

4 Result and Discussion

4.1 Result

The results of this study present an overview of the condition of household waste management by communities in Gorontalo City, with a main focus on two important aspects, namely the types of waste produced every day and people's behavior in disposing of waste. These two focuses are further analyzed based on three social dimensions, namely gender, age, and education level of the respondents, to understand how social background affects the way individuals and households treat the waste they produce.

4.1.1 Types of waste produced every day

(1) Gender * type of daily waste

The results of the crosstabulation analysis on Table 1 show that there is a difference in the distribution of the types of waste generated daily between male and female respondents in Gorontalo City. In general, the most common type of waste generated by both groups is organic waste, such as food waste and leaves, with a proportion of 70.4% in men and 74.6% in women. This shows that biodegradable household waste still dominates the daily waste composition in the community, regardless of gender.

Table 1. Gender * type of daily waste crosstabulation

			Types of Daily Waste								Total
			Organic Waste (Food Waste, Leaves)	Plastic	Paper/ Carton	Metal/ Glass	Wood Powder	Tiras	Mixed	Pampers	
Gender	Male	Count	107	40	3	1	0	0	1	0	152
		Expected Count	111	36.5	2.7	0.4	0.4	0.4	0.4	0.4	152
		% of Total	26.80%	10%	0.80%	0.30%	0%	0.00%	0.30%	0.%	38%
	Female	Count	185	56	4	0	1	1	0	1	248
		Expected Count	181	59.5	4.3	0.6	0.6	0.6	0.6	0.6	248
		% of Total	46.30%	14%	1%	0.%	0.3%	0.3%	0%	0.3%	62%
Total	Count	292	96	7	1	1	1	1	1	400	
	Expected Count	292	96	7	1	1	1	1	1	400	
	% of Total	73%	24%	1.8%	0.3%	0.3%	0.3%	0.3%	0.3%	100%	

Source: Data analysis, 2025.

However, there is a notable difference in the type of plastic waste, which tends to be reported more by men (26.3%) compared to women (22.6%). Meanwhile, paper/cardboard waste is generated more by men (2.0%) compared to women (1.6%). Although the percentage difference looks small, it indicates a variation in daily consumption activities based on gender, where men tend to generate more waste from packaging or single-use items.

(2) Age category * type of daily waste

The cross-tabulation results on Table 2 show that organic waste (such as food waste and leaves) is the most dominant type of waste generated by all age categories in Gorontalo City, with a total proportion of 73% of all respondents.

Table 2. Age category * type of daily waste crosstabulation

		Types of Daily Waste								Total
		Organic Waste (Food Waste, Leaves)	Plastic	Paper/ Carton	Metal/ Glass	Wood Powder	Tiras	Mixed	Pampers	
Age category	Count	4	5	0	0	0	0	0	0	9
	Teenager Expected Count	6.6	2.2	0.2	0	0	0	0	0	9
	% of Total	1.00%	1.30%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.30%
	Count	167	56	4	1	1	0	1	1	231
	Adult Expected Count	168.6	55.4	4	0.6	0.6	0.6	0.6	0.6	231
	% of Total	41.80%	14.00%	1.00%	0.30%	0.30%	0.00%	0.30%	0.30%	57.80%
	Count	95	25	3	0	0	1	0	0	124
	Pre elderly Expected Count	90.5	29.8	2.2	0.3	0.3	0.3	0.3	0.3	124
	% of Total	23.80%	6.30%	0.80%	0.00%	0.00%	0.30%	0.00%	0.00%	31.00%
	Count	26	10	0	0	0	0	0	0	36
	Elderly Expected Count	26.3	8.6	0.6	0.1	0.1	0.1	0.1	0.1	36
	% of Total	6.50%	2.50%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	9.00%
Total	Count	292	96	7	1	1	1	1	1	400
	Expected Count	292	96	7	1	1	1	1	1	400
	% of Total	73.00%	24.00%	1.80%	0.30%	0.30%	0.30%	0.30%	0.30%	100.00%

Source: Data analysis, 2025.

The adult group is the largest contributor to organic waste, accounting for 41.8% of the total respondents. This indicates that individuals in the productive age category tend to produce more organic waste, most likely due to their involvement in household activities and daily food consumption. Followed by the pre-elderly group at 23.8%, the elderly at 6.5%, and the least is the teenagers, only 1%. The very small number of the teenage group could indicate limited domestic activities or their lack of role in household waste management.

Plastic waste was the second highest, accounting for 24% of the total. Again, adults occupied the largest portion (14%), followed by pre-elderly (6.3%), elderly (2.5%), and teenagers (1.3%). This pattern is consistent with the organic waste type, reflecting that the productive age group is also dominant in the use and disposal of plastic. Other waste types such as paper/cardboard, metal/glass, sawdust, blinds, mixtures, and pampers show a very low proportion across all age groups, at only 0.3%–1.8% overall. However, pampers, mixtures, and waste slightly appear in the adult and pre-elderly categories, which could be related to household needs or more diverse activities.

From this data, it can be concluded that the adult age category is the largest contributor to all types of waste, especially organic and plastic waste. This indicates that waste management strategies in Gorontalo City need to be directed more towards productive age groups, either through education, provision of waste sorting facilities, or household-based waste reduction campaigns.

(3) Education * type of daily waste

Cross tabulation analysis on Table 3 shows that there is a variation in the type of waste generated daily by the people of Gorontalo City based on their education level. In general, the most dominant type of waste across all education levels is organic waste, such as food waste and leaves, with a combined total of 73% of all respondents.

Table 3. Education * type of daily waste crosstabulation

			Types of Daily Waste								Total
			Organic Waste (Food Waste, Leaves)	Plastic	Paper/ Carton	Metal/ Glass	Wood Powder	Tiras	Mixed	Pampers	
Education	SD	Count	39	22	1	0	0	0	1	0	63
		Expected Count	46	15.1	1.1	0.2	0.2	0.2	0.2	0.2	63
		% of Total	9.80%	5.50%	0.30%	0.00%	0.00%	0.00%	0.30%	0.00%	15.80%
	SMP	Count	30	17	1	0	1	0	0	0	49
		Expected Count	35.8	11.8	0.9	0.1	0.1	0.1	0.1	0.1	49
		% of Total	7.50%	4.30%	0.30%	0.00%	0.30%	0.00%	0.00%	0.00%	12.30%
	High School	Count	155	43	4	1	0	1	0	0	204
		Expected Count	148.9	49	3.6	0.5	0.5	0.5	0.5	0.5	204
		% of Total	38.80%	10.80%	1.00%	0.30%	0.00%	0.30%	0.00%	0.00%	51.00%
	D3/S1	Count	61	14	1	0	0	0	0	1	77
		Expected Count	56.2	18.5	1.3	0.2	0.2	0.2	0.2	0.2	77
		% of Total	15.30%	3.50%	0.30%	0.00%	0.00%	0.00%	0.00%	0.30%	19.30%
	Master's Degree	Count	7	0	0	0	0	0	0	0	7
		Expected Count	5.1	1.7	0.1	0	0	0	0	0	7
		% of Total	1.80%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	1.80%
Total			Count	292	96	7	1	1	1	1	400
			Expected Count	292	96	7	1	1	1	1	400
			% of Total	73.00%	24.00%	1.80%	0.30%	0.30%	0.30%	0.30%	100.00%

Note: SD, Sekolah Dasar (Elementary School); SMP, Sekolah Menengah Pertama (Junior High School); D3, Diploma 3; S1, Strata 1 (Bachelor's Degree). Source: Data analysis, 2025.

Respondents with high school education constituted the largest group in generating organic waste (38.8% of the total), followed by D3/S1 graduates (15.3%), elementary school (9.8%), and junior high school (7.5%). Meanwhile, respondents with S2/S3 background were the least (1.8%), indicating that the higher the education level, the smaller the proportion in the total number of respondents-although this is more reflective of the sample distribution rather than waste management patterns. In the case of plastic waste, the distribution pattern shows that high school graduates are also the largest contributor (10.8%), followed by elementary school (5.5%), junior high school (4.3%), D3/S1 (3.5%), and none from S2/S3. This shows that plastic is still a significant type of waste at all levels of education, although its use tends to be higher at the secondary level (high school).

(4) Chi-Square analysis results (types of waste produced)

The Chi-Square test results on Table 4 show that there is no significant relationship between the main social characteristics of respondents and the types of waste produced. The gender variable is not related to the composition of household waste ($\chi^2(7) = 5.948$, $p = 0.546$), nor is the age group ($\chi^2(35) = 26.955$, $p = 0.833$). Furthermore, education level also did not show a significant relationship with the type of waste produced ($\chi^2(28) = 31.232$, $p = 0.307$), although the linear-by-linear association test indicated a weak tendency for respondents with higher education to produce a more homogeneous waste composition ($p = 0.023$).

Table 4. Chi-Square analysis results (types of waste produced)

Independent Variabel	Dependent Variabel	χ^2	df	p-Value	Description
Gender	Types of waste	5.948	7	0.546	Not significant
Age	Types of waste	26.955	35	0.833	Not significantn
Education	Types of waste	31.232	28	0.307	Not significant, there is a weak trend (p -trend = 0.023)

Source: Data analysis, 2025.

Overall, these findings indicate that the composition of household waste in Gorontalo City is relatively uniform and is not significantly influenced by individual sociodemographic factors. This pattern is reasonable in the context of urban households, where the type of waste is more influenced by collective household consumption patterns than by individual differences.

4.1.2 Waste disposal behavior

(1) Gender * garbage disposal behavior

Based on the analysis, Table 5 shows result there is a significant difference between men and women in terms of waste disposal patterns. In general, women dominate in all types of waste disposal behavior, as the proportion of female respondents in this survey is greater (62%) than that of men (38%).

Table 5. Gender * garbage disposal behavior

		Garbage Disposal Behavior					Total
		Disposed of in Public Trash	Dumped into River/Empty Land	Taken by DLH Officers Gorontalo City	Sold to Collectors/Garbage Banks	Burned	
Gender	Count	49	8	72	1	22	152
	Male						
	Expected Count	48.3	9.5	70.7	0.8	22.8	152
	% of Total	12.30%	2.00%	18.00%	0.30%	5.50%	38.00%
	Count	78	17	114	1	38	248
	Female						
Total	Expected Count	78.7	15.5	115.3	1.2	37.2	248
	% of Total	19.50%	4.30%	28.50%	0.30%	9.50%	62.00%
	Count	127	25	186	2	60	400
	Expected Count	127	25	186	2	60	400
	% of Total	31.80%	6.30%	46.50%	0.50%	15.00%	100.00%

Note: DLH, Dinas Lingkungan Hidup (Environmental Agency). Source: Data analysis, 2025.

The most common method of disposal in both genders is waste collection by officials of the Gorontalo City Environmental Agency, with a proportion of 28.5% in women and 18% in men. This shows that both men and women relatively rely on official waste collection services, although women seem to be more active or better served in this regard. Disposal to public bins came second, with 19.5% of women and 12.3% of men engaging in this practice. Women again recorded higher rates, indicating a possibly more active role in the daily management of household waste. Waste burning, as an environmentally unfriendly practice, is still practiced by some respondents, especially by women (9.5%), compared to men (5.5%). This shows the tendency of women, perhaps due to greater domestic responsibilities, to solve waste problems in a practical but less sustainable way. Dumping waste into rivers or vacant land, although generally low (6.3%), is done more by women (4.3%) than men (2.0%). This behavior needs special attention as it contributes to environmental pollution.

(2) Category age * garbage disposal behavior

The results of the analysis on the Table 6 show that overall, the majority of respondents (46.5%) dispose of their waste by having it picked up by the officers of the Gorontalo City Environmental Agency, which reflects the functioning of formal waste collection services in the city. This pattern is consistent across age groups, especially in the adult category, which accounts for 25.3% of the total respondents. The pre-elderly group is also quite high, with 14.5%, while adolescents and the elderly are relatively smaller, at 1.5% and 5.3% respectively.

A total of 31.8% of respondents chose to throw their waste into public bins, making it the second most common method. Here, the adult group was again the biggest contributor (18.3%), followed by the pre-elderly (10.8%), elderly (2.0%), and teenagers (0.8%). This shows that the productive age group makes more use of the waste disposal facilities available in their neighborhood. Waste burning is still a significant practice, practiced by 15% of respondents overall. This practice is most common among adults (9.5%) and pre-elderly (4.3%), which may be due to limited access or preference for traditional ways of managing household waste. A small proportion of respondents (6.3%) still dispose of waste into rivers or bare soil, which is an environmentally unfriendly disposal practice. This practice is most commonly practiced by adults (4.3%), followed by pre-elderly (1.5%) and elderly (0.5%). No adolescents were reported to engage in this behavior, although the overall number of adolescent respondents was very small indeed.

Table 6. Category age * garbage disposal behavior

			Garbage Disposal Behavior				Total	
			Disposed of in Public Trash	Dumped into River/Empty Land	Taken by DLH Officers Gorontalo City	Sold to Collectors/Garbage Banks		Burned
Age category	Teenjater	Count	3	0	6	0	0	9
		Expected Count	2.9	0.6	4.2	0	1.3	9
		% of Total Count	0.80%	0.00%	1.50%	0.00%	0.00%	2.30%
	Adult	Count	73	17	101	2	38	231
		Expected Count	73.3	14.4	107.4	1.2	34.7	231
		% of Total Count	18.30%	4.30%	25.30%	0.50%	9.50%	57.80%
	Pre elderly	Count	43	6	58	0	17	124
		Expected Count	39.4	7.8	57.7	0.6	18.6	124
		% of Total Count	10.80%	1.50%	14.50%	0.00%	4.30%	31.00%
	Elderly	Count	8	2	21	0	5	36
		Expected Count	11.4	2.3	16.7	0.2	5.4	36
		% of Total Count	2.00%	0.50%	5.30%	0.00%	1.30%	9.00%
Total	Count	127	25	186	2	60	400	
	Expected Count	127	25	186	2	60	400	
	% of Total	31.80%	6.30%	46.50%	0.50%	15.00%	100.00%	

Note: DLH, Dinas Lingkungan Hidup (Environmental Agency). Source: Data analysis, 2025.

(3) Education * garbage disposal behavior

Based on the Table 7, it can be seen that education level has a fairly clear relationship with waste disposal behavior. The higher the level of education, it seems that the greater the tendency of people to use official waste disposal services.

Table 7. Education * garbage disposal behavior

			Garbage Disposal Behavior				Total	
			Disposed of in Public Trash	Dumped into River/Empty Land	Taken by DLH Officers Gorontalo City	Sold to Collectors/Garbage Banks		Burned
Education	SD	Count	20	5	24	0	14	63
		Expected Count	20	3.9	29.3	0.3	9.5	63
		% of Total Count	5.00%	1.30%	6.00%	0.00%	3.50%	15.80%
	SMP	Count	16	4	16	0	13	49
		Expected Count	15.6	3.1	22.8	0.2	7.4	49
		% of Total Count	4.00%	1.00%	4.00%	0.00%	3.30%	12.30%
	High School	Count	68	15	96	2	23	204
		Expected Count	64.8	12.8	94.9	1	30.6	204
		% of Total Count	17.00%	3.80%	24.00%	0.50%	5.80%	51.00%
	D3/S1	Count	21	1	45	0	10	77
		Expected Count	24.4	4.8	35.8	0.4	11.6	77
		% of Total Count	5.30%	0.30%	11.30%	0.00%	2.50%	19.30%
	Master's Degree	Count	2	0	5	0	0	7
		Expected Count	2.2	0.4	3.3	0	1.1	7
		% of Total Count	0.50%	0.00%	1.30%	0.00%	0.00%	1.80%
Total	Count	127	25	186	2	60	400	
	Expected Count	127	25	186	2	60	400	
	% of Total	31.80%	6.30%	46.50%	0.50%	15.00%	100.00%	

Note: DLH, Dinas Lingkungan Hidup (Environmental Agency); SD, Sekolah Dasar (Elementary School); SMP, Sekolah Menengah Pertama (Junior High School); D3, Diploma 3; S1, Strata 1 (Bachelor's Degree). Source: Data analysis, 2025.

Waste collection by Environmental Agency officers is the most chosen method by all education groups, with the largest proportion of high school graduates (24%) and D3/S1 (11.3%), and relatively small in the S2/S3 group (1.3%) because the number of respondents in this group is very small. This shows that people with middle and higher education tend to be more accessible or prefer to use government services in waste management. Disposal to public waste bins is also the top choice in almost all groups, especially by those with senior high school (17%) and primary school (5%) education, indicating that awareness to dispose of waste in its proper place is fairly evenly distributed across all education levels. Waste burning is still widely practiced, especially by high school graduates (5.8%), elementary school graduates (3.5%), and D3/S1 graduates (2.5%). This indicates that despite high levels of education, environmentally unfriendly practices still occur, possibly due to service availability or habit. Dumping into rivers or vacant lots, although small in number, is still of particular concern, practiced by 1.3% of primary school-educated respondents, 1% of junior high school-educated respondents, and very little in the other groups. This indicates that this behavior is more common in groups with lower education.

(4) Results of Chi-Square analysis on waste disposal behavior

The Chi-Square test results (Table 8) show that there is no significant relationship between the main social characteristics of respondents and waste disposal behavior. No significant differences were found based on gender ($\chi^2(4) = 0.608$, $p = 0.962$), age group ($\chi^2(20) = 19.641$, $p = 0.481$), or education level ($\chi^2(16) = 23.000$, $p = 0.114$). Thus, the waste disposal patterns of the people of Gorontalo City are homogeneous, with all social groups exhibiting relatively similar disposal behaviors. These findings indicate that waste management behavior is more influenced by household habits and access to services than by individual social characteristics.

Table 8. Results of Chi Square analysis on waste disposal behavior

Social Variables	Measured Variables	χ^2	df	p-Value	Description
Gender	Waste disposal behavior	0.608	4	0.962	Not significant
Age group	Waste disposal behavior	19.641	20	0.481	Not significantn
Level of education	Waste disposal behavior	23.000	16	0.114	Not significantn

Source: Data analysis, 2025.

4.2 Discussion

This study reveals significant new findings in understanding the waste disposal behavior of Gorontalo City residents, particularly in relation to sociodemographic variables such as age, gender, and education level. Through a quantitative approach with crosstab analysis of 400 respondents from 50 villages in 9 sub-districts, this study provides a detailed picture of daily waste disposal patterns that have not previously been explored in depth at the local level. One of the key findings is the dominance of the adult age group in generating organic and plastic waste, as well as their tendency to use the official waste collection service from the Environmental Agency. However, the practice of burning waste and dumping it in unauthorized locations still occurs, especially among people with low education levels. This is in line with the research findings of Maryati et al. [19], who also found that based on waste management knowledge, the community in Gorontalo City still uses methods of collection around their homes and burning waste, as well as disposal in rivers and irrigation systems to reduce waste accumulation [19]. This indicates that although access to formal services is available, there is still a gap in environmentally friendly waste management behavior.

These findings enrich the existing literature, which primarily focuses on policy and infrastructure aspects of waste management. Socialization is not enough; the government must ensure that community participation remains moderate rather than comprehensive [20]. Meanwhile, the waste transportation capacity should cover the total daily waste volume to be maximized [21, 22]. In this context, this research offers a new contribution by highlighting the importance of a demographic-based approach in designing waste management strategies. By understanding that waste disposal behavior is influenced by age and education level, more targeted and effective interventions can be designed, such as environmental education programs tailored to specific target groups.

In addition, this study also emphasizes the need to improve waste management infrastructure and facilities, such as Waste Management Reduce, Reuse, Recycle (TPS3R), which currently only two out of ten TPS3R are functioning properly in Gorontalo City. This shows that besides behavioral aspects, structural factors also play an important role in the effectiveness of waste management [23]. However, it is interesting that Evans et al. [24] found differently that age, education and gender were not associated with waste management behavior. While some studies support the results of this research, as found by Conti et al. [25] that there is a correlation between education level, knowledge and attitude towards waste management behavior. The same thing was also revealed by Sulistyorini et al. [26] that littering behavior was influenced by a low level of education by 57%.

In addition to descriptive findings, inferential analysis using the Chi-square test was conducted to determine whether the differences in patterns in the cross table were statistically significant. Although descriptive analysis showed small variations in waste disposal behavior between social groups, such as the tendency for women to burn

waste slightly more often, as also found in previous studies, the results of the Chi-square analysis in this study did not support these differences statistically. In other words, the variation seen in the contingency table was not strong enough to be generalized ($p > 0.05$). This suggests that the pattern is likely contextual and influenced by domestic work patterns in Gorontalo households, where women still play a dominant role in managing daily household activities, including household waste management.

This difference can also be explained by the characteristics of Gorontalo city, which generally has relatively good access to waste transportation services in most areas, so that burning is more related to environmental habits and the availability of local facilities than to gender preferences. Thus, these findings do not contradict previous literature but rather show that the relationship between gender and waste behavior in Gorontalo is more operational and situational than structural. These inferential findings are in line with environmental behavior theory, which emphasizes the role of social context and access to facilities in shaping behavior [27, 28].

The statistical insignificance of these findings actually indicates the homogeneity of waste management behavior at the household level, which confirms that the design of waste management policies and technologies in Gorontalo City should be directed more towards a universal approach based on public services, rather than the segmentation of specific social groups.

With no significant influence of social factors on waste disposal behavior, the design of the waste management information system for Gorontalo City should adopt a universal service approach. Features such as transportation schedules, reporting of illegal waste collection points, access to waste bank information, and online education are more appropriately developed as general functions for all citizens, rather than features segmented by demographic categories. Although the descriptive patterns show minor variations, the statistical insignificance needs to be viewed in the context of the broad category structure and the existence of several cells with low frequencies. Therefore, these findings emphasize the need for a general population-based policy approach and further studies that include structural variables such as service access, distance to waste collection points, and the quality of environmental education.

5 Conclusions

This study concludes that household waste management patterns in Gorontalo City are relatively homogeneous across all socio-demographic groups. Household waste is dominated by organic and plastic waste, while government transportation services and public waste facilities are the most commonly used disposal methods. Although descriptive analysis shows minor variations between social groups, the results of the Chi-square test show no significant relationship between gender, age group, or education level and the type of waste and disposal behavior ($p > 0.05$). This indicates that the apparent differences are not statistically significant enough to be generalized.

These findings suggest that individual social factors are not the main determinants of waste disposal behavior, and that waste management practices are more influenced by household habits and access to waste facilities. Thus, a universal approach to policy and management strategies is more appropriate than interventions targeting specific demographic groups. This study provides an important empirical basis for the pre-design stage of the Gorontalo City waste management information system, where the focus of system design needs to be directed at improving public services, ease of reporting, public education, and the integration of community-based waste facilities.

Further research is recommended to explore additional structural factors such as access to waste collection points, frequency of transportation services, the role of environmental institutions, and public awareness and digital literacy. By expanding the variables and methodological approaches, it is hoped that a more comprehensive waste information system can be designed to support effective, inclusive, and sustainable waste management in Gorontalo City.

Author Contributions

Conceptualization, S.M. and H.A.; methodology, H.A.; software, G.J.; validation, B.A., R.T.R.L.B., and A.E.; formal analysis, H.A.; investigation, S.M.; resources, A.E.; data curation, R.T.R.L.B.; writing—original draft preparation, H.A.; writing—review and editing, S.M.; visualization, H.A.; supervision, S.M.; project administration, B.A.; funding acquisition, H.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare that they have no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

In writing this article, the authors used AI applications such as ChatGPT to help find good synonym for words, and Grammarly to help improve sentence structure.

References

- [1] E. S. Sofiyah, I. W. K. Suryawan, A. Rahman, N. Ulhasanah, and M. M. Sari, "Socio-demographic and governance drivers of sanitation participation: Preliminary evidence from Gorontalo City, Indonesia," *Int. J. Mar. Eng. Innov. Res.*, vol. 10, no. 4, pp. 1244–1257, 2025. <https://doi.org/10.12962/J25481479.V10I4>
- [2] M. Nusi, Arman, S. Prahara, and D. A. Razak, "Implementation of waste management policy," *J. Indones. Law Policy Rev.*, vol. 6, no. 2, pp. 351–354, 2025. <https://doi.org/10.56371/JIRPL.V6I2.486>
- [3] A. Kaharu, S. Saman, M. F. Dunggio, and B. Idji, "Prevention and enhancement of housing quality and slum settlements of Biawu-Biawao, Gorontalo City," *Int. J. Innov. Sci. Res. Technol.*, vol. 9, no. 3, pp. 837–848, 2024. <https://doi.org/10.38124/IJISRT/IJISRT24MAR464>
- [4] M. Usman, T. Lamangida, and M. F. Akbar, "Implementation of waste management policy with 3R principles (Reduce, Reuse, Recycle) in the Gorontalo City," *J. Public Represent. Soc. Provision*, vol. 1, no. 1, pp. 29–37, 2021. <https://doi.org/10.55885/JPRSP.V1I1.36>
- [5] P. Unroj, P. Manomaivibool, and P. Dokmaingam, "Reducing energy consumption in waste management: A study on the recyclable waste bank approach," *Int. J. Energy Sect. Manag.*, 2026. <https://doi.org/10.1108/IJESM-03-2025-0006>
- [6] R. A. Damiti, F. Lihawa, and D. W. Baderan, "Evaluation of the functionality of 3R waste management facilities in Gorontalo Province," *J. Biol. Lingkung. Ind. Kesehat.*, vol. 12, no. 2, pp. 158–167, 2026. <https://doi.org/10.31289/BIOLINK.V12I2.16608>
- [7] Z. Mantali, "Identifying the implementation of three attributes of green city in Gorontalo City," *Sriwijaya J. Environ.*, vol. 8, no. 3, pp. 172–180, 2024. <https://doi.org/10.22135/SJE.2023.8.3.172-180>
- [8] N. Ulhasanah, A. Rahman, E. S. Sofiyah, I. W. K. Suryawan, and M. M. Sari, "Preliminary analysis of governance, social norms, and behavioral determinants of household participation in marine-linked waste management in Gorontalo Province, Indonesia," *Int. J. Mar. Eng. Innov. Res.*, vol. 10, no. 4, pp. 1133–1145, 2025. <https://doi.org/10.12962/J25481479.V10I4>
- [9] M. Azizah, A. Tahir, and Z. Abdussamad, "Evaluation of waste management policy in Talumelito landfill, Gorontalo Province, Indonesia," *Asian J. Adv. Res. Rep.*, vol. 18, no. 12, pp. 100–110, 2024. <https://doi.org/10.9734/AJARR/2024/V18I12809>
- [10] I. Y. Wulansari, T. H. Siagian, and E. Pasaribu, "Estimation of households that manages their domestic waste in not eco-friendly ways in municipal level in Indonesia using EBLUP Fay-Herriot model," *IOP Conf. Ser. Earth Environ. Sci.*, vol. 519, no. 1, 2020. <https://doi.org/10.1088/1755-1315/519/1/012057>
- [11] Y. E. Prasetyo, B. Setiawan, R. H. Alam, S. Seftiani, T. I. Datau, I. Butolo, and S. D. J. Biahimo, "Plastic waste management: Opportunities and challenges in implementing circular economy in Gorontalo," *Proc. Int. Conf. Multidiscip. Res. Sustain. Innov.*, vol. 1, pp. 65–72, 2024. <https://doi.org/10.31098/ICMRSI.V1I1.784>
- [12] M. Z. Alim, "The role of waste banks in realizing good environmental governance in Bandarlampung City," *J. Gov. Account. Stud.*, vol. 3, no. 1, pp. 51–61, 2024. <https://doi.org/10.35912/jgas.v3i1.1804>
- [13] M. Satori, R. Amaranti, and Y. Srirejeki, "Sustainability of waste bank and contribution of waste management," *IOP Conf. Ser. Mater. Sci. Eng.*, vol. 830, no. 3, p. 032077, 2020. <https://doi.org/10.1088/1757-899X/830/3/032077>
- [14] M. S. Han, Y. Zhang, and C. Liu, "How do two facets of social media interaction shape waste sorting behaviour? Empirical evidence from Beijing, China," *Technol. Forecast. Soc. Change*, vol. 210, p. 123850, 2025. <https://doi.org/10.1016/J.TECHFORE.2024.123850>
- [15] H. Eshete, A. Desalegn, and F. Tigu, "Knowledge, attitudes and practices on household solid waste management and associated factors in Gelemso town, Ethiopia," *PLoS One*, vol. 18, no. 2, p. e0278181, 2025. <https://doi.org/10.1371/journal.pone.0278181>
- [16] F. Wang, Q. Liu, and X. Yin, "Exploring influencing factors of consumers' wasted household appliances disposal behavior considering product heterogeneity in China," *J. Clean. Prod.*, vol. 418, p. 138065, 2023. <https://doi.org/10.1016/j.jclepro.2023.138065>

- [17] M. Sharma, D. Kaushal, S. Joshi, A. Kumar, and S. Luthra, "Electronic waste disposal behavioral intention of millennials: A moderating role of electronic word of mouth (eWOM) and perceived usage of online collection portal," *J Clean Prod.*, vol. 447, p. 141121, 2024. <https://doi.org/10.1016/j.jclepro.2024.141121>
- [18] T. Poyai, P. Pongpunpurt, U. Leknoi, P. Painmanakul, and N. Chawaloephonsiya, "Plastic waste management in urban areas: Key takeaways from the 'Send Plastic Home' project in Bangkok, Thailand," *Process Saf. Environ. Prot.*, vol. 190, pp. 1222–1232, 2024. <https://doi.org/10.1016/j.psep.2024.08.013>
- [19] S. Maryati, B. Ahaliki, R. T. R. Bau, A. Engelen, G. Juniarti, and I. Indrawan, "Assessing the community practices and perceptions on waste sorting in urban Indonesia: A case study in Gorontalo City," *Geographica: Sci. Educ. J.*, vol. 7, no. 1, pp. 35–46, 2025. <https://doi.org/10.31327/GSEJ.V7I1.2495>
- [20] S. Garg and P. Arora, "Waste management challenges and opportunities: A holistic approach for sustainable development," *Int. J. Environ. Sci. Technol.*, vol. 22, no. 14, pp. 14 753–14 770, 2025. <https://doi.org/10.1007/S13762-025-06653-5>
- [21] S. Shabarudin and H. Thamrin, "Factors contributing to the failure of environmental governance in waste management at the regional level," *Unifikasi: J. Ilmu Huk.*, vol. 12, no. 1, pp. 36–47, 2025. <https://doi.org/10.25134/UNIFIKASI.V12I01.1023>
- [22] D. Junus, N. I. Harun, and S. Napir, "Adopting sustainable environmental policy based on quadruple helix model in Gorontalo city, Indonesia," *Otoritas: J. Ilmu Pemerintah.*, vol. 14, no. 3, pp. 573–588, 2024. <https://doi.org/10.26618/OJIP.V14I3.12764>
- [23] D. I. Hussain, D. A. Elomri, D. L. Kerbache, and D. A. E. Omri, "Smart city solutions: Comparative analysis of waste management models in IoT-enabled environments using multiagent simulation," *Sustain. Cities Soc.*, vol. 103, p. 105247, 2024. <https://doi.org/10.1016/J.SCS.2024.105247>
- [24] A. K. Evans, N. L. Saw, C. E. Woods, L. M. Vidano, S. E. Blumenfeld, R. K. Lam, E. K. Chu, C. Reading, and M. Shamloo, "Impact of high-fat diet on cognitive behavior and central and systemic inflammation with aging and sex differences in mice," *Brain Behav. Immun.*, vol. 118, pp. 334–354, 2024. <https://doi.org/10.1016/J.BBI.2024.02.025>
- [25] A. Conti, E. Viottini, R. I. Commoreto, C. Piovan, B. Martin, B. Albanesi, M. Clari, V. Dimonte, and S. Campagna, "The effectiveness of educational interventions in improving waste management knowledge, attitudes, and practices among healthcare workers: A systematic review and meta-analysis," *Sustainability*, vol. 16, no. 9, p. 3513, 2024. <https://doi.org/10.3390/SU16093513/S1>
- [26] L. Sulistyorini, M. H. Abualressh, M. A. A. A. E. Regal, S. M. Elias, R. Azizah, L. Lutpiatina, N. D. Arfiani, M. A. Rizaldi, R. Z. Jannah, and M. B. Santanunurti, "Analysis of community knowledge and behavior towards plastic waste pollution control in the coastal area of Banyuwangi Regency, Indonesia," *Environ. Sustain. Indic.*, vol. 27, p. 100773, 2025. <https://doi.org/10.1016/J.INDIC.2025.100773>
- [27] S. Gul and W. Ahmed, "Enhancing the theory of planned behavior with perceived consumer effectiveness and environmental concern towards pro-environmental purchase intentions for eco-friendly apparel: A review article," *Bull. Bus. Econ.*, vol. 13, no. 1, pp. 784–791, 2024. <https://doi.org/10.61506/01.00270>
- [28] E. Crocetti, F. Albarello, W. Meeus, and M. Rubini, "Identities: A developmental social-psychological perspective," *Eur. Rev. Soc. Psychol.*, vol. 34, no. 1, pp. 161–201, 2023. <https://doi.org/10.1080/10463283.2022.2104987>